

Question				Marking details						Marks Available																									
										AO1		AO2		AO3		Total		Maths		Prac															
3	(a)	(i)		Purple White and Aa aa (1) Aa Aa aa aa (1) correct completion of punnett square							2			2																					
				Null hypothesis: There is no <u>significant</u> difference between the observed and expected numbers/ratios (1) <table><tr><td>Phenotype</td><td>O</td><td>E (1)</td><td>O-E</td><td>(O-E)²</td><td>$\frac{(O-E)^2}{E}$ (1)</td></tr><tr><td>white</td><td>32</td><td>25</td><td>7</td><td>49</td><td>1.96</td></tr><tr><td>purple</td><td>18</td><td>25</td><td>-7</td><td>49</td><td>1.96</td></tr></table> Chi ² = 3.92 (1)						Phenotype	O	E (1)	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$ (1)	white	32	25	7	49	1.96	purple	18	25	-7	49	1.96		4			4			4
Phenotype	O	E (1)	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$ (1)																														
white	32	25	7	49	1.96																														
purple	18	25	-7	49	1.96																														
		(ii)																																	
		(iii)		Critical value is 3.84 (1) because chi ² value {is greater than/to the right of critical value (at 0.05)} (1) reject null hypothesis (1) deviation is not due to chance/there is a significant difference between O and E(1) ECF from (ii)									4	4		4	4																		
	(b)	(i)		<table><tr><td></td><td>AB</td><td>Ab</td><td>aB</td><td>ab</td></tr><tr><td>aB</td><td>AaBB</td><td>AaBb</td><td>aaBB</td><td>aaBb</td></tr><tr><td>ab</td><td>AaBb</td><td>Aabb</td><td>aaBb</td><td>aabb</td></tr></table> Gametes (1) Correct completion (1) [from their gametes] Phenotype ratio 3 purple : 5 white (1) [from table]						AB	Ab	aB	ab	aB	AaBB	AaBb	aaBB	aaBb	ab	AaBb	Aabb	aaBb	aabb		3			3							
				AB	Ab	aB	ab																												
aB	AaBB	AaBb	aaBB	aaBb																															
ab	AaBb	Aabb	aaBb	aabb																															
		(ii)		Hypothesis 2 and The chi squared value is {to the left/less than} the critical value/no <u>significant</u> difference between <u>O</u> and <u>E</u> /probability is between {0.8-0.9/above 0.8/above 5%}/due to chance alone									1	1		1																			